

Continuous Probability

UChicago Political Science Math Camp

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2025

Continuous Probability

We can expand our understanding of probability to include infinite outcomes

We know that a random variable can model discrete outcomes: i.e., flipping a coin, rolling a dice

What if we took a dice with one face for every real number $\mathbb{R} \in [0, 1]$?

PDFs

A PDF extends a PMF to include a continuous domain, such that

$$f(x) = \Pr[X = x]$$

Where f returns a real number for every input of x . A PDF returns the probability density of a certain outcome of a random variable.

Probability Axioms

Recall our probability axioms:

1. $\Pr[a] \geq 0 \forall a \in A$
2. $\sum_{a \in A} \Pr[a] = 1$
3. For disjoint outcomes, $\Pr[\bigcup_{i=1}^k a_i] = \sum_{i=1}^k \Pr[a_i]$

How might we apply these to continuous probability distributions?

PDF Warnings

A few things become questionable with PDFs:

1. PDFs do not need to be continuous nor differentiable.
2. PDFs can be defined for negative inputs (RV outcomes), just not negative outputs (probabilities)
3. PDFs do not need to be injective nor surjective

Conceptualize: what is this PDF telling me?

PDF Bounds

It's not uncommon to see a PDF take a form such as:

$$f(x) = 2x$$

But, the condition $\int_{\mathbb{R}} f(x)dx = 1$ still must hold. We then create **bounds** which allow this:

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$$\begin{aligned}\int_a^b 2x dx = 1 &\implies x^2 \Big|_a^b = 1 \\ \implies b^2 - a^2 = 1 &\implies b = \sqrt{a^2 + 1}\end{aligned}$$

More generally, we must choose a, b such that $F(b) - F(a) = 1$.

CDFs

Say for some PDF $f(X)$, we want to know the likelihood that x is larger than some X . We can use the **cumulative distribution function**:

$$F(x) = \Pr[X \leq x] = \int_{-\infty}^x f(s)ds$$

We use FTC to obtain this: $\int_a^b f(x)dx = F(b) - F(a)$, for $a = -\infty$, and some arbitrary s

Expanded CDFs

This FTOC definition allows us to specify the probability of an RV falling within a given range. For example,

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We can then use our **disjoint probability axiom** to combine multiple domains:

$$\Pr[x \in [a, b] \text{ OR } [c, d]] = \int_a^b f(x)dx + \int_c^d f(x)dx$$

CDF Foundations

What do our first two axioms imply about continuous CDFs?

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- ▶ Weakly increasing: $F(x) \geq F(x') \forall x > x'$
- ▶ Comprehensive: $F(\infty) = 1$

Joint Distributions

Joint distributions are an easy extension from discrete probability:

$$p(x, y) = P(X = x, Y = y)$$

$$\implies f(x, y); P((X, Y) \in A) = \int \int_A f(x, y) dx dy$$

Probability represented by volume under the joint PDF

Marginal Distributions

A **marginal distribution**, denoted $f_X(x)$, reflects the distribution of one variable, holding the other variable(s) constant.

Then, we can represent by integrating **w.r.t. the other variable!**

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx$$

Joint PDFs

You cannot always derive a joint PDF, but: if two events are independent, their joint PDF is the product of their marginal PDFs

Example: Let $f(x) = e^{-x}$ and $f(y) = N(\bar{x}, \sigma)$. What is $f(x, y)$?

Conditional Probability

To evaluate the **conditional probability** of a continuous RV, we hold the other variable constant:

$$\Pr[y|x] = f_{y|x}(y|x) = \frac{f(x,y)}{f_X(x)}$$

Uniform Distribution

The uniform distribution takes a slightly different form for continuous probability:

$$f(x) = \frac{1}{b - a}$$

for uniformity over the domain $[a, b]$.

Expanding Expectation

Recall that for a discrete RV, $E[X] = \sum_{x \in X} \Pr[x] \cdot x$

How can we expand this to an infinite cardinality of x ?

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$$E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

Unfortunately, you can't isolate out $\int_{\mathbb{R}} f(x)dx = 1$ here

An Integration Corollary:

The sum rule lends that

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$$

Suppose $f(x)$ is only defined on $[a, b]$ and therefore takes a value of 0 for all other inputs. Then,

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c 0dx$$

$$\implies \int_a^b f(x)dx$$

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Much of probability calls for infinite integrals, but we can often isolate a well-defined domain.

Median

The median defines a value such that half of the mass is above that threshold, and half is below.

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$$\text{Median}(X) \equiv x \text{ s.t. } \int^x f(X) dX = \frac{1}{2}$$